

Which Way Did It Blow

84-355 Democracy's Data | The election wind map, and what an arrow promises

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Everything here runs as given. Your job is to read what comes out and say what it means.

Read `wind-map-brief.Rmd` first. This lab runs the same steps with the code exposed.

The claim an arrow makes

“Miami-Dade swung eleven points.”

Every wind map is thousands of sentences like that at once. Each one says: **this is one place, I observed it twice, and here is the difference.**

That is three claims, and any of them can be false while the arrow still gets drawn.

Step 1 — The encoding, fixed once

```
ff <- read.csv(file.path(D, "facts.csv"))
FV <- setNames(ff$value, ff$key)
F <- function(k) { z <- FV[[k]]; n <- suppressWarnings(as.numeric(z))
  if (is.na(n)) z else n }

data.frame(
  channel = c("direction", "direction", "length", "length", "thickness"),
  rule = c("straight up = no change",
    paste0(F("deg_per_point"), " degrees clockwise per point toward the Republican"),
    paste0(F("km_per_point_us"), " km per point (national map)"),
    paste0("stops growing past ", F("len_cap_pts"), " points"),
    "square root of votes cast"))
```

Channel	Rule
direction	straight up = no change
direction	5 degrees clockwise per point toward the Republican
length	18 km per point (national map)

Channel	Rule
length	stops growing past 15 points
thickness	square root of votes cast

Direction and length carry the same number twice. That redundancy is the point: the map survives being printed in black and white.

The scale is a declared constant, not something fitted to the data on the page. A map that rescales its arrows to fill its own frame is a picture of the frame.

Step 2 — The national map, and what it is made of

```
us <- read.csv(file.path(D, "wind_us.csv"),
               colClasses = c(county_fips = "character")) # leading zeros!
us <- us[us$in_frame == "TRUE" | us$in_frame == TRUE, ]
c(counties      = nrow(us),
  median_swing  = round(median(us$swing), 2),
  share_toward_R = round(100 * mean(us$swing > 0), 1),
  largest_R_shift = round(max(us$swing), 1),
  largest_D_shift = round(min(us$swing), 1))
```

```
#>      counties  median_swing  share_toward_R  largest_R_shift  largest_D_shift
#>      3100.00         3.06         88.50         28.10         -9.20
```

Read `county_fips` as `character`. Read as a number, 01001 becomes 1001 and Alabama’s counties join to nothing. A sibling lab lost 1,549 of 4,489 identifiers to exactly this.

Step 3 — Who published this?

There is **no federal source for county-level presidential returns**. The Federal Election Commission publishes official results by state and by congressional district. County returns are published by 51 separate state election offices in 51 formats.

So the national file used above is somebody’s compilation. Its own README says the 2024 figures were scraped from a television network’s results pages and that the numbers **“are not authoritative.”**

That does not make it useless. It makes it a file you have to audit.

Step 4 — Audit the units

```
c20 <- read.csv("../data-sources/data/pres2020_counties.csv",
                colClasses = c(county_fips = "character"))
c24 <- read.csv("../data-sources/data/pres2024_counties.csv",
                colClasses = c(county_fips = "character"))
c(rows_2020 = nrow(c20),
  rows_2024 = nrow(c24),
```

```
in_both    = length(intersect(c20$county_fips, c24$county_fips)),
only_2020  = length(setdiff(c20$county_fips, c24$county_fips)),
only_2024  = length(setdiff(c24$county_fips, c20$county_fips))
```

```
#> rows_2020 rows_2024   in_both only_2020 only_2024
#>      3152      3160      3104       48       56
```

Eight more rows in 2024 than in 2020. **The country did not gain eight counties.** Take the non-matching codes apart:

```
aud <- read.csv(file.path(D, "provenance_audit.csv"),
                 colClasses = c(fips = "character"))
table(sub("^[A-Za-z ]+\\.\\*", "\\1", aud$reason), aud$appears_in)
```

```
#>
#>                2020 only 2024 only
#> Alaska                40      40
#> Connecticut           8       9
#> District of Columbia  0       7
```

Three different things, only one of which is a change on the ground:

- **Alaska** does not report presidential votes by county at all. Both files carry 40 State House Districts under **different made-up FIPS codes** — 02901-02940 in one file, 02001-02040 in the other. Same kind of unit, different arbitrary labels. And Alaska redrew those districts after the 2020 census, so matching by district number would compare different ground anyway.
- **The District of Columbia** is one row in 2020 and eight wards in 2024. Nothing moved; somebody reported it finer.
- **Connecticut** is the real unit change. The Census Bureau retired its eight counties and adopted **nine planning regions** as county equivalents, with new codes, effective for 2022 and after.

Step 5 — The failure that does not show up in a set difference

Alaska and Connecticut fail loudly: the codes do not match and the rows drop out. Watch what the District of Columbia does.

```
rbind(c20[c20$county_fips == "11001", 1:6],
      c24[c24$county_fips == "11001", 1:6])
```

State name	County fips	County name	Votes dem	Votes gop	Total votes
District of Columbia	11001	District of Columbia	317323	18586	344356
District of Columbia	11001	Ward 1	36882	2056	40463

Same FIPS. Different place. The join succeeds and an arrow gets drawn comparing a whole city to one eighth of itself.

```
mg <- function(d, g) 100 * (g - d) / (g + d)
naive <- mg(c24$votes_dem[c24$county_fips == "11001"],
           c24$votes_gop[c24$county_fips == "11001"]) -
  mg(c20$votes_dem[c20$county_fips == "11001"],
     c20$votes_gop[c20$county_fips == "11001"])
dc24 <- c24[substr(c24$county_fips, 1, 2) == "11", ]
true <- mg(sum(dc24$votes_dem), sum(dc24$votes_gop)) -
  mg(c20$votes_dem[c20$county_fips == "11001"],
     c20$votes_gop[c20$county_fips == "11001"])
c(naive_swing = round(naive, 2), true_swing = round(true, 2),
  error = round(true - naive, 2))
```

```
#> naive_swing true_swing error
#>      -0.51      2.30     2.81
```

The arrow reverses. Naively the District moved toward the Democrats; summed properly it moved toward the Republicans, by more than the median county moved at all.

And both joins return the same number of rows:

```
c(naive_join_rows = nrow(merge(c20, c24, by = "county_fips")),
  harmonised_rows = as.integer(F("matched_after_harmonise")))
```

```
#> naive_join_rows harmonised_rows
#>           3104           3104
```

The row count cannot warn you. Nothing about a successful join tells you whether the two sides mean the same thing.

Step 6 — What harmonising cost

```
data.frame(
  state = c("Alaska", "District of Columbia", "Connecticut"),
  action = c("dropped all 40 units",
             "eight wards summed back to one District",
             "dropped"),
  cost = c("no presidential arrows for Alaska",
            "none; this is a repair",
            paste0(format(F("ct_votes_lost"), big.mark = ","), " votes")))
```

State	Action	Cost
Alaska	dropped all 40 units	no presidential arrows for Alaska
District of Columbia	eight wards summed back to one District	none; this is a repair
Connecticut	dropped	1,824,280 votes

Two of the three repairs are deletions. **A wind map cannot draw an arrow for a place that did not survive**, and a hole you have counted is better than an arrow you have invented.

Step 7 — One arrow per county, or one per voter?

```
v <- us$votes_dem_24 + us$votes_gop_24
agg <- function(d, g) 100 * (sum(g) - sum(d)) / (sum(g) + sum(d))
c(median_county_swing = round(median(us$swing), 2),
  mean_county_swing   = round(mean(us$swing), 2),
  national_vote_swing = round(agg(us$votes_dem_24, us$votes_gop_24) -
                                agg(us$votes_dem_20, us$votes_gop_20), 2),
  largest_county       = max(v), smallest_county = min(v))

#> median_county_swing mean_county_swing national_vote_swing largest_county
#>                3.06                3.40                6.04        3606971.00
#> smallest_county
#>                96.00
```

The counties moved about three points. The country moved about six. Both are correct and they answer different questions.

One arrow per county gives the smallest county in the file the same visual weight as Los Angeles, which cast roughly 37,573 times as many votes. Thickness compensates a little. It does not compensate much.

That is an editorial choice, not a technical one, and this lab has made it.

Step 8 — Georgia, from the state's own record

```
ga <- read.csv(file.path(D, "wind_ga.csv"), colClasses = c(fips = "character"))
c(counties      = nrow(ga),
  margin_2020   = round(F("ga_state_margin_20"), 2),
  margin_2024   = round(F("ga_state_margin_24"), 2),
  statewide_swing = round(F("ga_state_swing"), 2),
  median_county  = round(median(ga$swing), 2),
  share_toward_R = round(100 * mean(ga$swing > 0), 1))

#> counties margin_2020 margin_2024 statewide_swing median_county
#>      159.00      -0.26         2.21         2.47         2.96
#> share_toward_R
#>       81.80
```

Both years are the **Georgia Secretary of State's own record** — the 2020 general election archive and the state's 2024 results export. Georgia has had 159 counties since 1945, so nothing was recoded, nothing was renamed, and no arrow here can be fictional.

Same encoding, different provenance. That is the only difference between this map and the national one, and it is the whole difference.

Step 9 — Why not precincts?

Georgia has 2,701 precincts — seventeen times the resolution. A companion lab has already carried the 2020 votes onto the 2024 precinct boundaries, so the boundary problem is solved. All that remains is attaching the 2024 returns by precinct **name**.

```
read.csv(file.path(D, "precinct_join_ladder.csv"))
```

Rung	Matched	Unmatched 2020	Pct of 2020
1. exact name, uppercased	2198	494	81.6
2. strip leading ballot-order number	2353	339	87.4
3. strip trailing (CITY) tag	2354	338	87.4
4. strip all punctuation and spaces	2446	246	90.9

```
head(read.csv(file.path(D, "precinct_join_residual.csv")), 6)
```

County	Unmatched	Total 2020	Ballots 2020	Ballots unmatched
CHATHAM	86	86	131490	131490
COLUMBIA	17	46	79248	31614
DOUGLAS	12	25	68272	33624
CARROLL	11	28	53702	23963
LOWNDES	11	18	45810	17538
DOUGHERTY	10	25	35023	11267

The ladder stalls at 90.9%, and the loss is not scattered: **CHATHAM County loses all 86 of its precincts**, because the returns identify precincts by codes like 1-05C and the boundary file identifies them by the name of the polling place.

A precinct map would have a hole where Savannah is, and **the hole would look like a quiet neighbourhood**. That is why Step 8 is drawn at county level.

Step 10 — The 2026 arrow, and the office problem

The midterm is **Tuesday, 3 November 2026**, during this semester. The plan is to draw the same map in class once returns exist.

A 2024-to-2026 arrow would compare a presidential race to a midterm: different office, different electorate, different turnout. How much does that matter?

Georgia's November 2020 ballot carried a presidential race **and** a US Senate race. Same voters, same day, same piece of paper. Any disagreement is a pure office effect with **zero** turnout difference.

```
og <- read.csv(file.path(D, "office_gap.csv"))
c(precincts      = nrow(og),
  statewide_pres  = round(F("og_state_pres"), 2),
  statewide_senate = round(F("og_state_sen"), 2),
  office_gap      = round(F("og_state_gap"), 2),
  median_precinct_gap = round(median(abs(og$gap)), 2),
  pct_over_2_points = round(100 * mean(abs(og$gap) > 2), 1),
  georgia_2020_2024_swing = round(F("ga_state_swing"), 2))
```

```
#>           precincts      statewide_pres      statewide_senate
#>           2641.00           -0.23           1.84
#>           office_gap      median_precinct_gap      pct_over_2_points
#>           2.08           1.63           42.00
#> georgia_2020_2024_swing
#>           2.47
```

The office effect alone is about two points — the same size as the entire real shift Georgia experienced across two presidential elections. And that is the best case, with the electorate held exactly fixed.

So this lab does not draw a 2024-to-2026 arrow. The class map is Georgia's 2022 governor's race against its 2026 governor's race: same office, both midterms, same 159 counties, same publisher.

Step 11 — The half that is already built

```
g22 <- read.csv(file.path(D, "wind_ga_counties.csv"),
                colClasses = c(fips = "character"))
c(counties      = nrow(g22),
  dem_2022      = F("ga22_dem"),
  gop_2022      = F("ga22_gop"),
  margin_2022   = round(F("ga22_margin"), 2),
  counties_with_a_2026_result = sum(!is.na(g22$margin_to)))
```

```
#>                counties                dem_2022
#>                159.00                1813673.00
#>                gop_2022                margin_2022
#>                2111572.00                7.59
#> counties_with_a_2026_result
#>                0.00
```

November 8, 2022 - General/Special Election, snapshot 2025-01-08, straight from results.sos.ga.gov. Every county has a tail. None has a head yet.

What has to happen on class day

One file changes. `data/wind_input_2026.csv` has ten columns and 159 rows; two of the columns are empty:

```
str(read.csv(file.path(D, "wind_input_2026.csv"),
              colClasses = c(unit_id = "character")))
```

```
#> 'data.frame': 159 obs. of 10 variables:
#> $ unit_id      : chr "13001" "13003" "13005" "13007" ...
#> $ unit_name    : chr "Appling" "Atkinson" "Bacon" "Baker" ...
#> $ state        : chr "Georgia" "Georgia" "Georgia" "Georgia" ...
#> $ office       : chr "Governor" "Governor" "Governor" "Governor" ...
#> $ year_from    : int 2022 2022 2022 2022 2022 2022 2022 2022 2022 2022 ...
#> $ year_to      : int 2026 2026 2026 2026 2026 2026 2026 2026 2026 2026 ...
#> $ votes_dem_from: int 1131 467 393 500 6913 607 7309 8137 1680 772 ...
#> $ votes_gop_from: int 5552 1767 3312 755 7987 6651 21833 31528 3412 5209 ...
#> $ votes_dem_to  : logi NA NA NA NA NA NA ...
#> $ votes_gop_to  : logi NA NA NA NA NA NA ...
```

Column	Meaning	Rule
<code>unit_id</code>	5-digit county FIPS	character , leading zeros kept
<code>unit_name</code>	county name	must match the baseline
<code>office</code>	the office compared	identical in both years, or stop
<code>year_from, year_to</code>	2022, 2026	both midterms
<code>votes_dem_from,</code> <code>votes_gop_from</code>	2022 governor	already filled
<code>votes_dem_to, votes_gop_to</code>	2026 governor	the two empty columns

The refresh is one call:

```
Rscript -e 'source("data/build-data.R"); build_2026()'
```

which fetches `export-2026NovGen.json` from the Secretary of State, fills the two columns, recomputes every swing, and rewrites `wind_ga_counties.csv`. If the state has not posted the export, edit the two columns by hand from any source and call `build_2026(from_stub = TRUE)`.

What will not be knowable

A month after 3 November 2026, expect **unofficial** returns. Certification deadlines run into December in many states; close races will still be open; precinct-level returns rarely appear this fast. **The map drawn in class will be provisional and should be labelled so.** Every wind map published on election night has this property and most do not admit it.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Weight the map by votes. Step 7 shows the counties moving about three points while the country moved about six. Redraw the national map with arrows sized by votes cast instead of one per county — or simply report the swing separately for the largest 100 counties and the smallest 1,000. Which map is the honest one, and honest about what?

Option B — Break the harmonisation on purpose. Re-run Step 5 for every state, not just the District of Columbia: for each state, compare the number of rows in the two files. Which other states change their reporting grain between elections, and would you have caught them without being told to look?

Option C — Audit Georgia against the national file. The national file has Georgia counties too, from a press compilation. Compare its Georgia numbers to the Secretary of State's in `wind_ga.csv`, county by county. Where do they disagree, by how much, and does the disagreement change any arrow's direction?

Option D — Argue with the encoding. The map draws margin change and nothing else. Build the turnout-change map instead, using `tot_20` and `tot_24` in `wind_ga.csv`, and say which of the two you would put on a front page.

Option E — Your own question. 3,100 national arrows, 159 Georgia arrows, and one that has not been drawn yet.

What this lab cannot tell you

- **Whether the national numbers are right.** No federal publisher exists and the compilation says of itself that it is not authoritative. Georgia's are the state's own record; the rest are not.
- **Anything about Alaska.** Two files, two arbitrary coding schemes, and redrawn districts in between.
- **What happened in Connecticut.** Its eight counties became nine planning regions and those votes are simply absent from the national map.
- **Whether a swing was caused by persuasion or by turnout.** Margin change cannot separate them. A county where nobody changed their mind and a third of one side stayed home gets the same arrow as a county that was persuaded.
- **How uncertain an arrow is.** There is no error bar on a wind map and no agreed way to draw one.

On the discussion board

By Friday, 11:59 p.m.

- Step 5 shows a join that succeeded, reported the same row count as the correct one, and produced an arrow pointing the wrong way. **What check would have caught it?** Be specific enough that someone could run it.
- Step 7 says one arrow per county is an editorial choice. Whose interests does that choice serve, and is the alternative any less of a choice?
- Step 10 measures a two-point gap between two offices on the *same ballots*. Given that, what would you need before you believed any cross-year swing number smaller than two points?

Where the data came from

- **Georgia returns, all three elections** — Georgia Secretary of State. 2020 general election archive at <https://sos.ga.gov/page/historical-elections-results>; 2022 and 2024 results exports at <https://results.sos.ga.gov/>, e.g. <https://results.sos.ga.gov/cdn/results/Georgia/export-2022NovGen.json> (fetched 2026-08-10, HTTP 200).
- **County population centroids** — U.S. Census Bureau, *Centers of Population by County: 2020*, https://www2.census.gov/geo/docs/reference/cenpop2020/county/CenPop2020_Mean_CO.txt (fetched 2026-08-10, HTTP 200, 3,221 rows).
- **National county presidential returns** — Tony McGovern, *US County Level Election Results 08-24*, as committed in `../data-sources/`. https://github.com/tonmcg/US_County_Level_Election_Results_08-24 **Scraped from news organisations; not authoritative, by its own account.**
- **Official federal returns** — Federal Election Commission, <https://www.fec.gov/introduction-campaign-finance/election-results-and-voting-information/> Published by state and congressional district, **not by county**, which is why this lab needs the compilation above.
- **The wind-map idiom** — Observable, *Plot: election wind map*. <https://observablehq.com/@observablehq/plot-election-wind-map>

The build is `data/build-data.R`. It records every URL, the fetch date and the row counts, and it contains `build_2026()`, the one function that has to run in November.